

Hot and cold

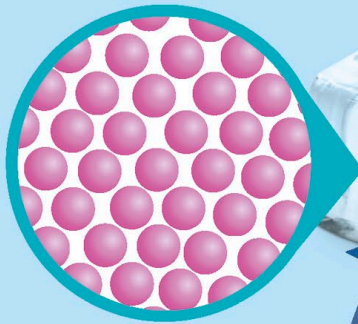
The temperature of an object depends on how the particles that make it up move around. If we add heat energy to a solid, its particles begin to move and it melts into a liquid. If we add more heat energy, the particles move even faster and the liquid evaporates into a gas.

Melting and boiling points

Different materials have different melting and boiling points. These are the temperatures at which a solid changes into a liquid, and then a gas. These processes can also work in reverse.

Solid

Solid water is called ice. The particles in a solid are in fixed positions. They vibrate, but they don't move around – this is why solids keep their shape.



Water vapour can turn straight into a solid by a process called deposition.

Solids can change straight into gases in a process called sublimation.

Gas

Water as a gas is called water vapour. Particles in a gas move quickly away from each other in all directions, filling the space they are in. When invisible water vapour starts to cool, it makes tiny droplets of liquid, which create steam.

